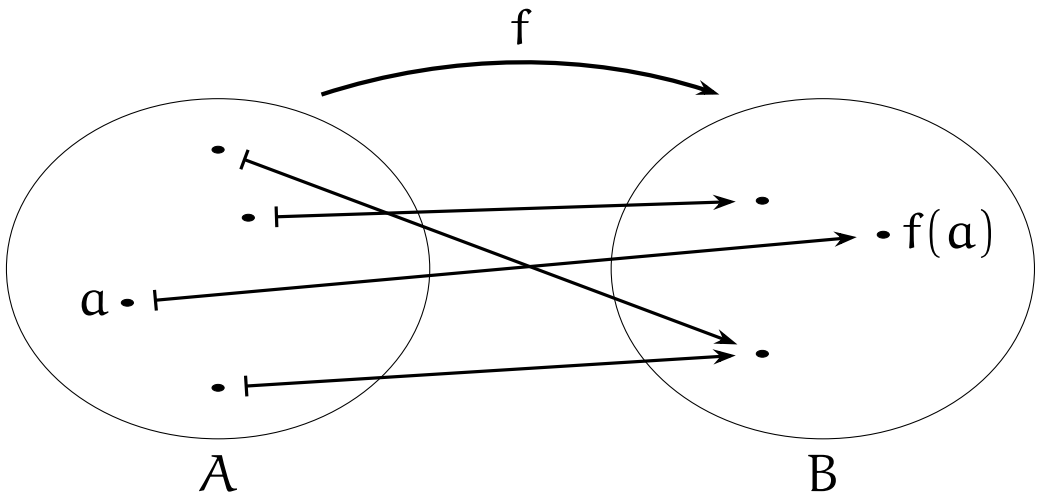
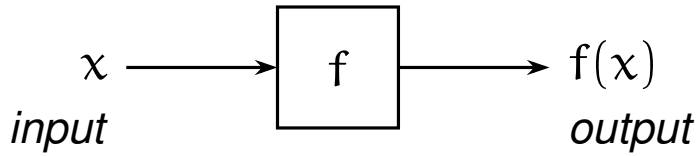
Functions

A ***function*** f from a set A to a set B, denoted $f : A \to B$, associates a *unique* value $f(a)$ of B to *each* value a of A, which we denote by $f : a \mapsto b$.



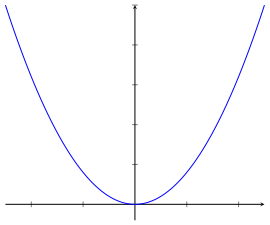
Thus, f is like a machine into which you feed a value x and it responds by outputting a value $f(x)$:



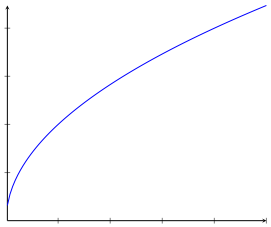
Functions

Functions are familiar in mathematics. For example:

- The squaring function $sq : \mathbb{R} \to \mathbb{R}$ takes a real value and squares it, and is typically written as $sq(x) = x^2$.



- The square root function $sqrt : \mathbb{R}^{\geq 0} \to \mathbb{R}^{\geq 0}$ takes a non-negative real number and returns its square root, and is typically written as $sqrt(x) = \sqrt{x}$.



- The function $floor : \mathbb{R} \to \mathbb{Z}$ takes a real number and return the largest integer which does not exceed the given real value; this is typically written $floor(x) = \lfloor x \rfloor$. Thus for example $floor(3.8)=3$ and $floor(-3.8)=-4$.

Functions

Example: Grades (as integer percentages) of the students in a class:

Archer	75	Evans	78	Parks	64
Byron	92	Farmer	46	Smith	59
Cole	64	Greene	68	Taylor	100
Davies	88	Lewis	54	Wells	78

Each person in the set $\text{Class} = \{\text{Archer}, \text{Byron}, \text{Cole}, \text{Davies}, \text{Evans}, \text{Farmer}, \text{Greene}, \text{Lewis}, \text{Parks}, \text{Smith}, \text{Taylor}, \text{Wells}\}$

is assigned a value from the set $\text{Marks} = \{0, 1, 2, 3, \ldots, 100\}$

This describes a function $\text{score} : \text{Class} \to \text{Marks}$

For example, $\text{score}(\text{Greene}) = 68$

- the function score maps $\text{Greene} \in \text{Class}$ to $68 \in \text{Marks}$
- that is, $\text{score}: \text{Greene} \mapsto 68$

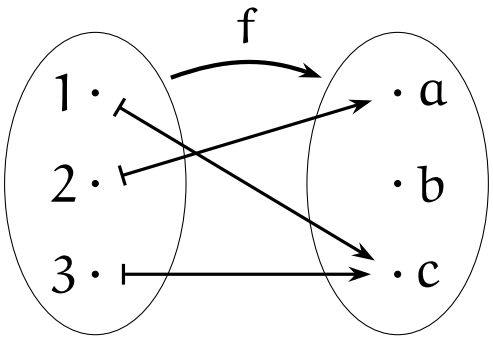
Domain, Codomain and Range

If $f: A \to B$, then

- A is called the ***domain*** of f. Thus, $\text{domain}(\text{score}) = \text{Class}$
- B is called the ***codomain*** of f. Thus, $\text{codomain}(\text{score}) = \text{Marks}$
- The ***range*** of f is the subset of the codomain consisting of the values that are produced by f: $\text{range}(f) = \{ f(a) : a \in A \}$

Thus, $\text{range}(\text{score}) = \{ 46, 54, 59, 64, 68, 75, 78, 88, 92, 100 \}$

Example:



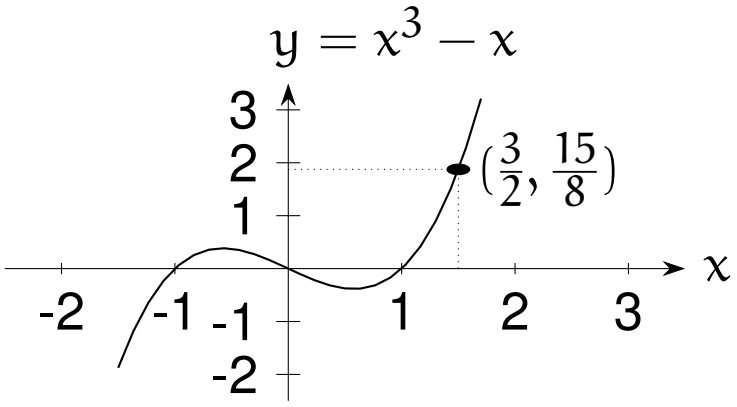
$\text{domain}(f) = \{1, 2, 3\}$

$\text{codomain}(f) = \{a, b, c\}$

$\text{range}(f) = \{a, c\}$

Graph of a function

Functions in mathematics are plotted as a graph, consisting of the collection of points "lying" on the graph. More generally, the ***graph*** of a function $f : A \to B$ is:



$$\text{graph}(f) = \{\, (a, b) \in A \times B : \; b = f(a) \,\}$$

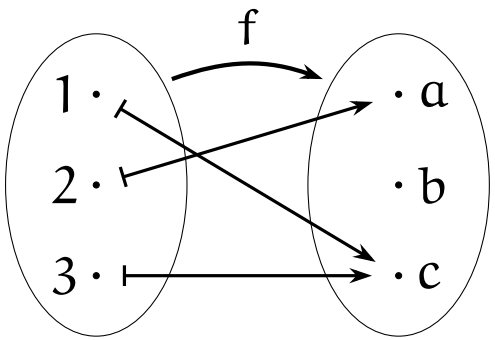
So for $f(x) = x^3 - x$: $\text{graph}(f) = \{\, (x\,,\, x^3 - x) : \; x \in \mathbb{R} \,\}$

and $\text{graph}(\text{score}) = \{\, (\text{Archer}, 75)\,,\ (\text{Byron}, 92)\,,\ (\text{Cole}, 64)\,,$
$(\text{Davies}, 88)\,,\ (\text{Evans}, 78)\,,\ (\text{Farmer}, 46)\,,$
$(\text{Greene}, 68)\,,\ (\text{Lewis}, 54)\,,\ (\text{Parks}, 64)\,,$
$(\text{Smith}, 59)\,,\ (\text{Taylor}, 100)\,,\ (\text{Wells}, 78)\ \}$

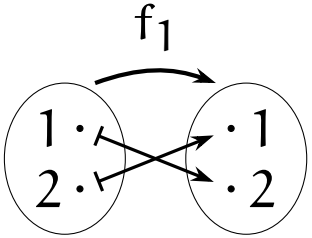
Two functions f and g are equal if, and only if, they share the same domain and codomain, and $\text{graph}(f) = \text{graph}(g)$.

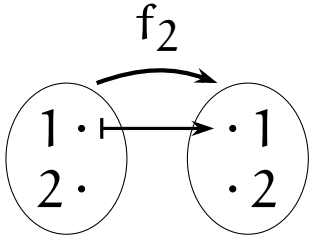
Functions and their Graphs

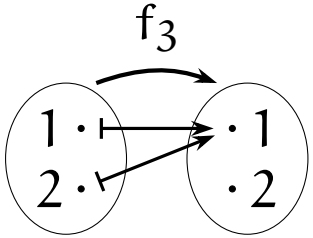
Exercise: What is the graph of the following function:

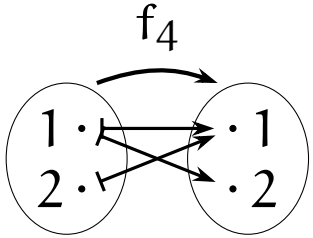


Exercise: Let $X = \{1, 2\}$. Which of the following define a function from X to X?









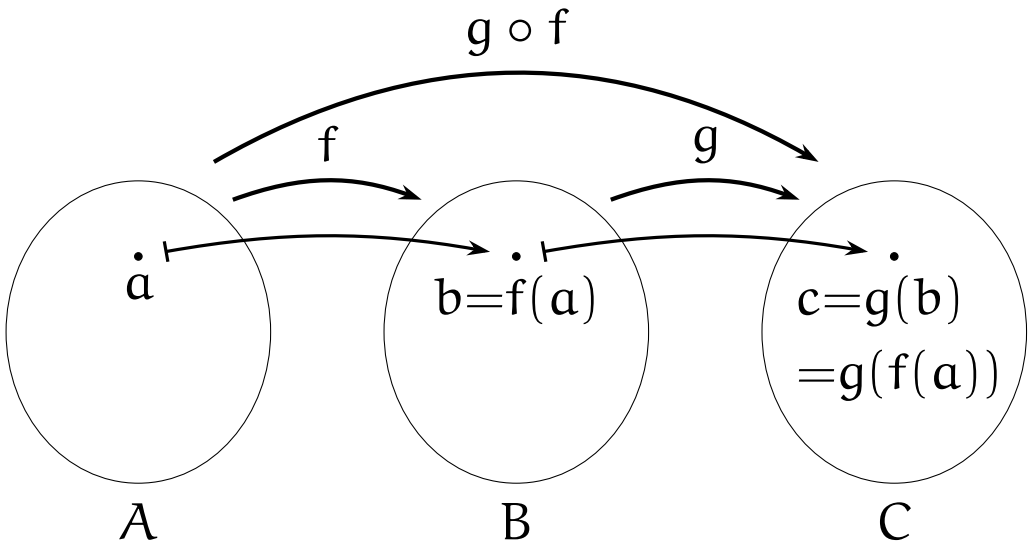
For those which are functions, determine their graphs.

Exercise: Find all other valid functions from X to X, and compute their graphs.

Composing Functions

We can ***compose*** two functions $f: A \to B$ and $g: B \to C$, – i.e., apply f followed by g – to get a new function $g \circ f : A \to C$.

Given $f: A \to B$ and $g: B \to C$, the ***composition*** of g and f is the function $g \circ f: A \to C$ defined by $(g \circ f)(x) = g(f(x))$.



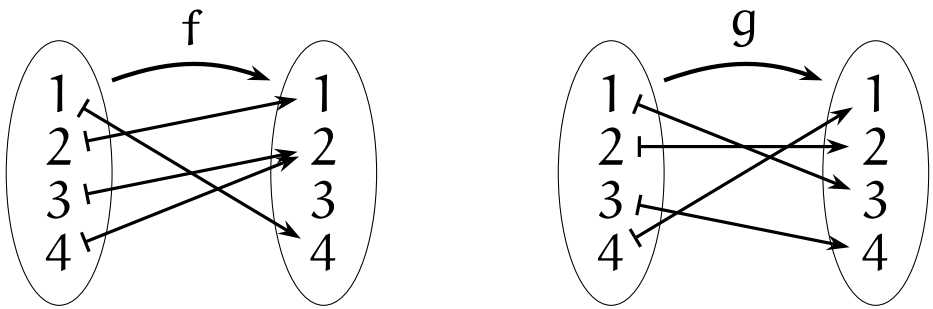
If $f: A \to A$ then we can form $f \circ f$, which we denote by f^2.

More generally, f^n applies the function f n times in a row:

$$f^0 = \mathrm{id}_A \qquad f^1 = f \qquad f^2 = f \circ f \qquad f^3 = f \circ f \circ f \qquad f^n = \underbrace{f \circ \cdots \circ f}_{n \text{ times}}$$

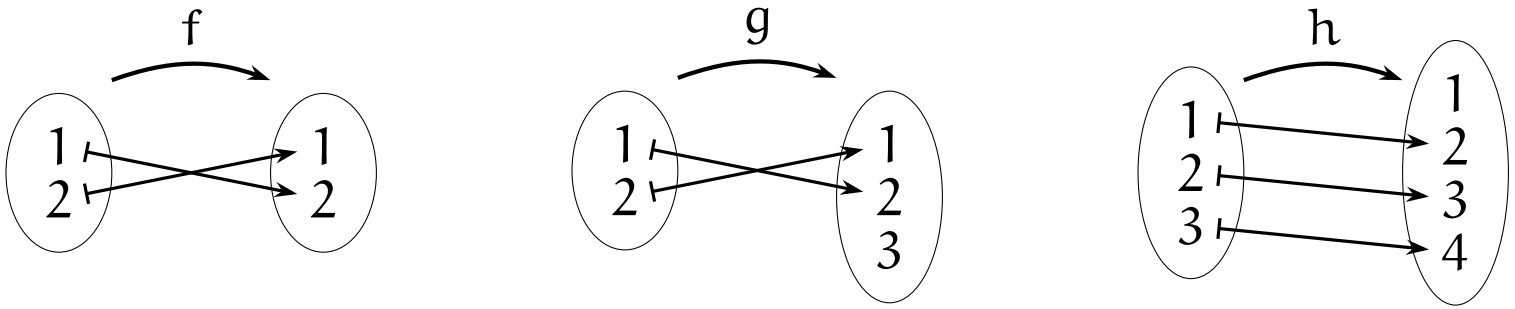
Composing Functions

Exercise: Let $A = \{1, 2, 3, 4\}$ and consider the functions $f, g : A \to A$



Find $f \circ g$, $g \circ f$, and f^2 (i.e., $f \circ f$).

Exercise: Consider the following functions:



What functions can be made from pairwise composing these functions?